System of Linear Equation

①

Important step

→ অনেকগুলো Equation দেয়া থাকবে, main kaj single variable এর answer, !! exchange position, so smallest first coefficient is at top.

(Echelon) (vanishing)

→ ~~First iteration~~, sob equation এর name = ($L_1, L_2, L_3, \ldots, L_n$)

~~L_1 eqn same রাখবো, L.~~

→ First iteration, $L_1' = L_1$

$L_2' = L_2 - L_1$

$L_3' = L_3 - L_1$

$L_n' = L_n - L_1$.

→ Second iteration, $L_1'' = L_1' = L_1$

$L_2'' = L_2'$

$L_3'' = L_3' - L_2'$

$L_4'' = L_4' - L_2'$

$L_n'' = L_n' - L_2'$

→ Third iteration, $L_1''' = L_1'' = L_1' = L_1$

$L_2''' = L_2'' = L_1'$

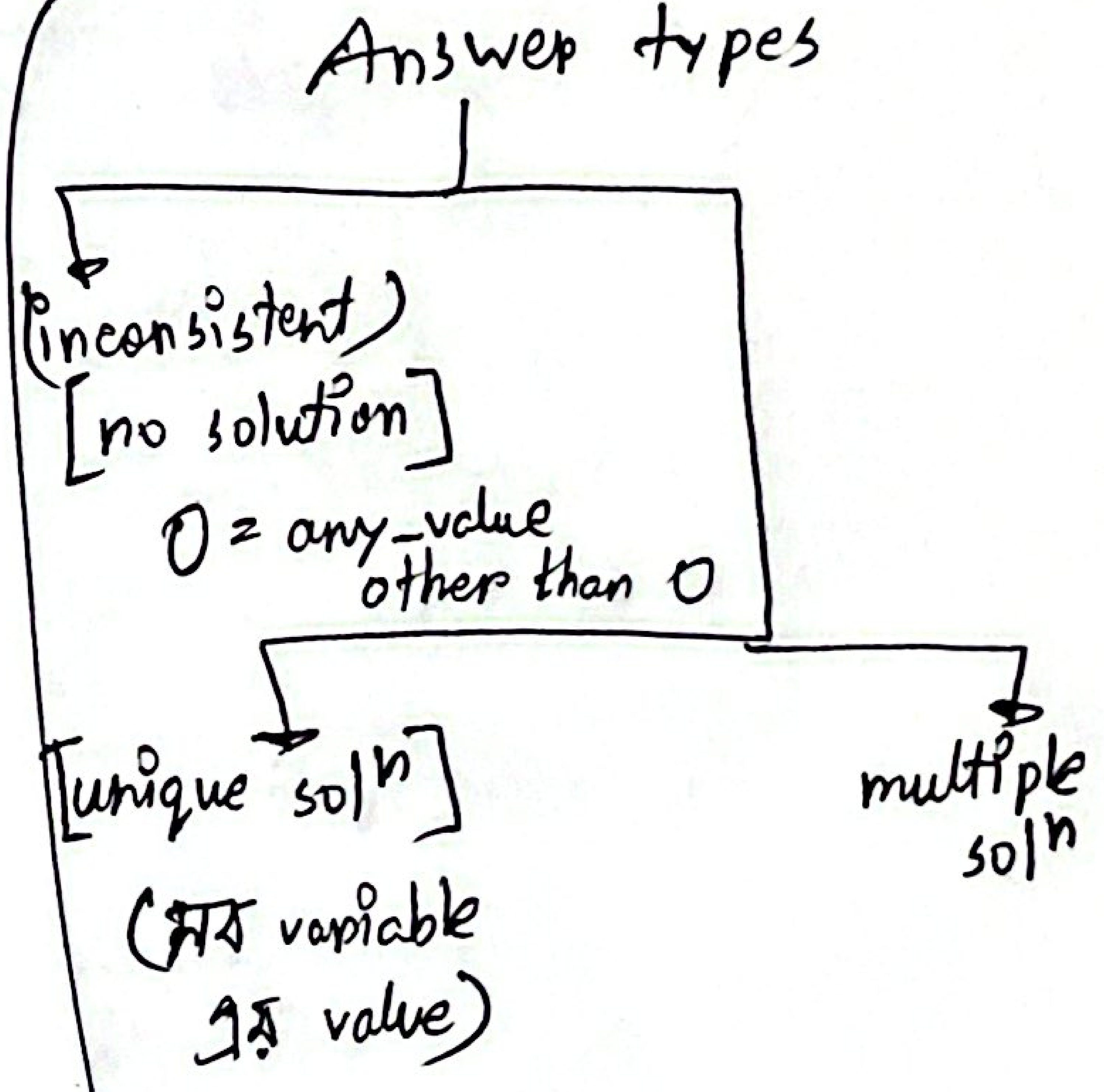
$L_3''' = L_3''$

$L_4''' = L_4'' - L_3''$

$L_5''' = L_5'' - L_3''$

$L_n''' = L_n'' - L_3''$

So on...



$$\boxed{\text{Question}}$$

swap

$$\boxed{\begin{array}{l} 3x_1 - x_2 + x_3 = -2 \\ x_1 + 5x_2 + 2x_3 = 6 \\ 2x_1 + 3x_2 + x_3 = 0 \end{array}} \Rightarrow \boxed{\begin{array}{ll} x_1 + 5x_2 + 2x_3 = 6 & \rightarrow L_1 \\ 3x_1 - x_2 + x_3 = -2 & \rightarrow L_2 \\ 2x_1 + 3x_2 + x_3 = 0 & \rightarrow L_3 \end{array}}$$

$L_1' \Rightarrow x_1 + 5x_2 + 2x_3 = 6$

$L_2' \Rightarrow 16x_2 + 5x_3 = 20$

$L_3' \Rightarrow 7x_2 + 3x_3 = 12$

$L_2' = L_2 \cancel{-L_1} - 3L_1$

$L_2 \Rightarrow \cancel{3x_1} - x_2 + x_3 = -2$

$3L_1 \Rightarrow \underset{(-)}{\cancel{3x_1}} \underset{(-)}{+} 15x_2 \underset{(-)}{+} 6x_3 \underset{(-)}{=} 18$

$L_2' \Rightarrow -16x_2 - 5x_3 = -20$

$L_3' = L_3 - 2L_1$

$L_3 = \cancel{2x_1} + 3x_2 + x_3 = 0$

$2L_1 = \underset{(-)}{\cancel{2x_1}} \underset{(-)}{+} 10x_2 \underset{(-)}{+} 4x_3 \underset{(-)}{=} 12$

$-7x_2 - 3x_3 = -12$

$L_1'' \Rightarrow x_1 + 5x_2 + 2x_3 = 6$

$L_2'' \Rightarrow 16x_2 + 5x_3 = 20$

$L_3'' \Rightarrow x_3 = 4$

$$\boxed{\begin{array}{l} x_3 = 4 \\ x_2 = 0 \\ x_1 = -2 \end{array}}$$

[Unique Solution]
(Ans)

$L_3'' = 16L_3' - 7L_2'$

$16L_3' \Rightarrow \cancel{112x_2} + 48x_3 = 192$

$7L_2' \Rightarrow \underset{(-)}{\cancel{112x_2}} \underset{(-)}{+} 35x_3 \underset{(-)}{=} 140$

$+13x_3 = 52$

$x_3 = 4$

Question

$$x_1 + 2x_2 + 3x_4 + x_5 = 0 \rightarrow L_1$$

$$2x_1 + 3x_2 + 3x_4 + x_5 = 0 \rightarrow L_2$$

$$x_1 + x_2 + x_3 + 2x_4 + x_5 = 0 \rightarrow L_3$$

$$3x_1 + 5x_2 + 6x_4 + 2x_5 = 0 \rightarrow L_4$$

$$2x_1 + 3x_2 + 2x_3 + 5x_4 + 2x_5 = 0 \rightarrow L_5$$

$L_1' \Rightarrow x_1 + 2x_2 + 3x_4 + x_5 = 0$

$L_2' \Rightarrow -x_2 - 3x_4 - x_5 = 0$

$L_3' \Rightarrow -x_2 + x_3 - x_4 = 0$

$L_4' \Rightarrow -x_2 - 3x_4 - x_5 = 0$

$L_5' \Rightarrow -x_2 + 2x_3 - x_4 = 0$

$L_2' \Rightarrow L_2 - 2L_1$

$L_2 \Rightarrow \cancel{2x_1} + 3x_2 + 3x_4 + x_5 = 0$

$2L_1 \Rightarrow \underset{(-)}{\cancel{2x_1}} \underset{(-)}{+ 4x_2} \underset{(-)}{+ 6x_4} \underset{(-)}{+ 2x_5} = \underset{(-)}{0}$

$-x_2 - 3x_4 - x_5 = 0$

$L_3' \Rightarrow L_3 - L_1$

$L_3 \Rightarrow \cancel{x_1} + x_2 + x_3 + 2x_4 + \cancel{x_5} = 0$

$L_1 \Rightarrow \underset{(-)}{\cancel{x_1}} \underset{(-)}{+ 2x_2} \underset{(-)}{+ 3x_4} \underset{(-)}{+ \cancel{x_5}} = \underset{(-)}{0}$

$-x_2 + x_3 - x_4 = 0$

$L_4' \Rightarrow L_4 - 3L_1$

$L_4 \Rightarrow \cancel{3x_1} + 5x_2 + 6x_4 + 2x_5 = 0$

$3L_1 \Rightarrow \underset{(-)}{\cancel{3x_1}} \underset{(-)}{+ 6x_2} \underset{(-)}{+ 9x_4} \underset{(-)}{+ 3x_5} = \underset{(-)}{0}$

$-x_2 - 3x_4 - x_5 = 0$

$L_5' = L_5 - 2L_1$

$L_5 \Rightarrow \cancel{2x_1} + 3x_2 + 2x_3 + 5x_4 + \cancel{2x_5} = 0$

$2L_1 \Rightarrow \underset{(-)}{\cancel{2x_1}} \underset{(-)}{+ 4x_2} \underset{(-)}{+ 6x_4} \underset{(-)}{+ \cancel{2x_5}} = \underset{(-)}{0}$

$-x_2 + 2x_3 - x_4 = 0$

$L_1'' \Rightarrow x_1 + 2x_2 + 3x_4 + x_5 = 0$

$L_2'' \Rightarrow -x_2 - 3x_4 - x_5 = 0$

$L_3'' \Rightarrow x_3 + 2x_4 + x_5 = 0$

$L_4'' \Rightarrow 0 = 0$

$L_5'' \Rightarrow 2x_3 + 2x_4 + x_5 = 0$

$L_3'' = L_3' - L_2'$

$L_3' \Rightarrow \cancel{-x_2} + x_3 - x_4 = 0$

$L_2' \Rightarrow \cancel{-x_2} - 3x_4 - x_5 = 0$

$(+) \quad (+) \quad (+) \quad (-)$

$x_3 + 2x_4 + x_5 = 0$

$L_4'' = L_4' - L_2'$

$L_4' \Rightarrow \cancel{-x_2} \cancel{-3x_4} \cancel{-x_5} = 0$

$L_2' \Rightarrow \cancel{-x_2} \cancel{-3x_4} \cancel{-x_5} = 0$

$(+) \quad (+) \quad (+) \quad (-)$

$0 = 0$

$L_5'' = L_5' - L_2'$

$L_5' \Rightarrow \cancel{-x_2} + 2x_3 - x_4 = 0$

$L_2' \Rightarrow \cancel{-x_2} - 3x_4 - x_5 = 0$

$(+) \quad (+) \quad (+) \quad (-)$

$2x_3 + 2x_4 + x_5 = 0$

~~[No Solution]~~

$L_1''' \Rightarrow x_1 + 2x_2 + 3x_4 + x_5 = 0$

$L_2''' \Rightarrow -x_2 - 3x_4 - x_5 = 0$

$L_3''' \Rightarrow x_3 + 2x_4 + x_5 = 0$

$L_5''' \Rightarrow$ ~~$2x_3 + 2x_4 + x_5 = 0$~~ $-2x_4 - x_5 = 0$

$L_5''' = L_5'' - 2L_3''$

$L_5'' \Rightarrow \cancel{2x_3} + 2x_4 + x_5 = 0$

$2L_3'' \Rightarrow \cancel{2x_3} + 4x_4 + 2x_5 = 0$

$(-) \quad (-) \quad (-) \quad (-)$

$-2x_4 - x_5 = 0$

Let $x_4 = a$

$\Rightarrow x_5 = -2a$

$\Rightarrow x_2 = -3a + 2a$

$x_2 = -a$

~~$\Rightarrow x_3 = -2a - 2a$~~

~~$x_3 = \frac{-4a - 3a}{2}$~~

~~$\Rightarrow x_1 = -10a - 3a + 4a$~~

~~$x_1 = -9a - a$~~

$\Rightarrow x_3 = -2x_4 - x_5$

$= -2a + 2a$

$= 0$

$\Rightarrow x_1 = -2x_2 - 3x_4 - x_5$

$= +2a - 3a + 2a$

$= a$

Question

$2x_1 + 3x_2 + 5x_3 + x_4 = 3$

$3x_1 + 4x_2 + 2x_3 + 3x_4 = -2$

swap

$x_1 + 2x_2 + 8x_3 - x_4 = 8$

$7x_1 + 9x_2 + x_3 + 8x_4 = 0$

$\Rightarrow$

$x_1 + 2x_2 + 8x_3 - x_4 = 8 \longrightarrow L_1$

$3x_1 + 4x_2 + 2x_3 + 3x_4 = -2 \longrightarrow L_2$

$2x_1 + 3x_2 + 5x_3 + x_4 = 3 \longrightarrow L_3$

$7x_1 + 9x_2 + x_3 + 8x_4 = 0 \longrightarrow L_4$

$L_1' \Rightarrow x_1 + 2x_2 + 8x_3 - x_4 = 8$

$L_2' \Rightarrow -\cancel{x_2 - 11} \; - -x_2 - 11x_3 + 3x_4 = -13$

$L_3' \Rightarrow -x_2 - 11x_3 + 3x_4 = -13$

$L_4' \Rightarrow -5x_2 - 55x_3 + 15x_4 = -56$

$L_2' = L_2 - 3L_1$

$L_2 \Rightarrow \cancel{3x_1} + 4x_2 + 2x_3 + \cancel{3x_4} = -2$

$3L_1 \Rightarrow \underset{(-)}{\cancel{3x_1}} \underset{(-)}{+} 6x_2 \underset{(-)}{+} 24x_3 \underset{(+)}{-} \cancel{3x_4} \underset{(-)}{=} 24$

$-2x_2 - 22x_3 + 6x_4 = -26$

$L_3' = L_3 - 2L_1$

$L_3 \Rightarrow \cancel{2x_1} + 3x_2 + 5x_3 + x_4 = 3$

$2L_1 \Rightarrow \underset{(-)}{\cancel{2x_1}} \underset{(-)}{+} 4x_2 \underset{(-)}{+} 16x_3 \underset{(+)}{-} 2x_4 \underset{(-)}{=} 16$

$-x_2 - 11x_3 + 3x_4 = -13$

$L_4' = L_4 - 7L_1$

$L_4 \Rightarrow \cancel{7x_1} + 9x_2 + x_3 + 8x_4 = 0$

$7L_1 \Rightarrow \underset{(-)}{\cancel{7x_1}} \underset{(-)}{+} 14x_2 \underset{(-)}{+} 56x_3 \underset{(+)}{\mp} 7x_4 \underset{(-)}{=} 56$

$-5x_2 - 55x_3 + 15x_4 = -56$

$L_1'' \Rightarrow x_1 + 2x_2 + 8x_3 - x_4 = 8$

$L_2'' \Rightarrow \quad -x_2 - 11x_3 + 3x_4 = -13$

$L_3'' \Rightarrow \quad 0 = 0$

[no solution] (Ans)

$L_3'' \Rightarrow L_3' - L_2'$

$$
\begin{array}{l}
L_3' \Rightarrow \cancel{-x_2} - \cancel{11x_3} + \cancel{3x_4} = \cancel{-13} \\
L_2' \Rightarrow \underset{(+)}{\cancel{-x_2}} \; \underset{(+)}{\cancel{-11x_3}} \; \underset{(-)}{\cancel{+3x_4}} = \underset{(+)}{\cancel{-13}} \\
\hline
\qquad\qquad\qquad 0 = 0
\end{array}
$$